

Complete Lab 5: Keyword Spotting Using TinyML. After completing the software and hardware parts of the lab, [upload all required files to the GitHub repository you are using for this class](#).

Your GitHub repository should contain:

1. [Your completed Jupyter notebook file \(.ipynb\)](#)
2. [Your Arduino files/sketch used for deployment](#)
3. [A single PDF file containing your answers to the questions below](#)

[Submit the GitHub repository link on Canvas](#). Make sure the repository link is accessible to the course instruction team.

Questions to Answer in Your PDF

1. In the notebook, the audio waveform is transformed into time-frequency features before being passed to the keyword spotting model. What is the purpose of converting raw audio into spectrogram/MFCC-style features instead of directly using the raw time-domain waveform?

We want to avoid using time-domain data in ML data processing because we need all model input data to have the same dimensions/signal length. Unfortunately for us, some words take longer to say than others. In order to use time domain audio data we would need to pad short data and trim any data that is too long. The former is stupid and the latter is actively destructive towards potentially critical data.

Anyway the time-frequency domain means that the length of all signals can be exactly the same, while not destroying any of that precious data.

2. During full integer quantization, the notebook uses a representative dataset. What is the purpose of the representative dataset, and why is it important for creating an INT8 TensorFlow Lite model?

The representative dataset is used as a “guide” when quantizing down our model. When converting to INT8 we are stripping away a ton of information, so we include the RD in the process to calibrate it so it doesn’t get totally off the mark.

3. What does `audio_processor.get_data()` do? Refer to `input_data.py`.

It gets samples from the attached data set, and “applies transformations as needed” to said data. Returns transformed sample data and label indexes

4. In the final model export step, the TensorFlow Lite model is converted into a C/C++ byte array. What are `g_model` and `g_model_len`, and why do they need to be copied correctly into the Arduino source file?

`G_model` is a binary representation of the model, aka just the model itself. You need this because... otherwise you aren't running the correct model

`G_model_len` is the size of your model. If you have the wrong number here your computer will read your model wrong and mess everything up.

5. Please include a screenshot showing the correct identification of at least one keyword, either “**YES**” or “**NO**”, as displayed in the Arduino IDE Serial Monitor after deploying your KWS model on the Arduino Nano 33 BLE. The screenshot should be included in your submitted PDF.

```
Heard yes (205) @712944ms
Heard unknown (206) @713760ms
Heard unknown (204) @715792ms
Heard unknown (253) @717360ms
Heard unknown (206) @720032ms
Heard unknown (209) @721872ms
Heard unknown (228) @723424ms
Heard no (209) @725824ms
Heard unknown (205) @726832ms
```